ET-617 · ASSIGNMENT

# LEARNFLOW

An interactive learning platform that turns every learner interaction into clickstream data.

📖 Text · 🎬 Video · ✅ Quizzes · 🖱 Clickstream · 📊 Analytics · ⬇ CSV · Excel · JSON

THE BRIEF

# What the assignment asked for

- A learning app where users **log in as learners**
- Interactive content: **text, video, quizzes** (or a combination)
- Track **all user activity** — clicks, page views, video actions, quiz attempts…
- Store that **clickstream data**
- A standalone web/mobile app, any stack, using **Git**

Reference data format: a **Moodle logstore** export (Time · Event context · Component · Event name · Description · Origin · IP).

THE SOLUTION

# LearnFlow at a glance

**Learners & admins**

Email + password auth (bcrypt, token sessions). Learners see their own data; admins see everyone's.

**Three content types**

Rich reading, HTML5 video with live event capture, and server-graded multiple-choice quizzes.

**Full clickstream**

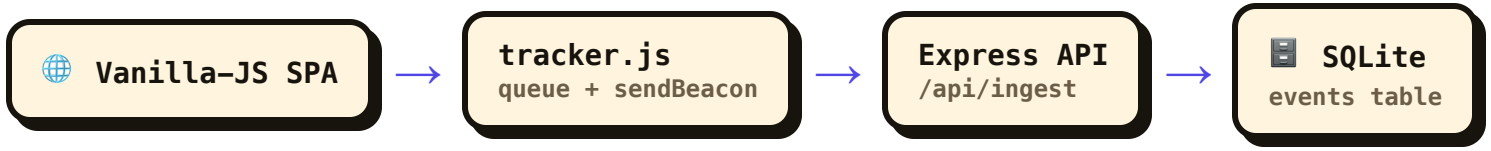
Clicks, scroll depth, time-on-page, video play/pause/seek/progress, quiz start/answer/submit, auth.

**Analytics + export**

Live dashboard with charts and a raw log, plus one-click export as CSV, Excel or JSON.

ARCHITECTURE

# One process, no build step




- Frontend loads plain `.js` / `.css` — runs with a single `npm start`.
- **better-sqlite3** — a real, indexed relational store with zero setup.
- Login & quiz-submit are logged **server-side** too, so the timeline can't be spoofed.

THE CORE

# How the clickstream flows

- **Capture** — `tracker.js` auto-records clicks, scroll & time-on-page, and exposes `Tracker.log()` for semantic events.
- **Batch** — events queue up and flush every few seconds, or via `sendBeacon` on tab-hide (no lost tail events).
- **Enrich** — the server stamps the real IP + authenticated user id.
- **Store** — one indexed `events` row per interaction.
- **Read back** — aggregates, raw log, and CSV/Excel/JSON export (scoped per role).

**Every event** carries both human-readable Moodle fields *and* machine fields (event_type, target_id, payload).

COVERAGE

# What gets tracked

| COMPONENT | EXAMPLE EVENTS |
| --- | --- |
| Authentication | registered · logged in · logged out |
| System | Dashboard / Course / Analytics viewed · exported |
| Lesson | Lesson viewed · marked complete |
| Video | played · paused · seeked · progress 25/50/75% · completed · volume · rate |
| Quiz | started · answer selected · attempt submitted |
| Interface | element clicked · scroll depth · time on page · theme changed |

DATA

# The `events` table = the clickstream

Moodle-style columns drive the export; rich columns drive analytics.

**Moodle columns**

Time · Event context · Component · Event name · Description · Origin · IP address

**Analytics columns**

user_id · session_id · event_type · target_type · target_id · url · payload(JSON)

Related tables: `users`, `sessions`, `courses`, `lessons`, `quiz_attempts`. Quiz answer keys are stripped before reaching the browser; grading is server-side.

LIVE DEMO

# Let's see it

- Log in → dashboard with a progress ring & live study timer
- Read a **text** lesson (scroll tracked)
- Watch a **video** — play / pause / seek logged live
- Take a **quiz** — instant colour-coded grading
- Open **My Analytics** → charts + raw log → **Export**

`npm install && npm run seed && npm start` → `localhost:3000`

WRAP-UP

# Everything the brief asked — and then some

✅ **Delivered**

Learner auth · text/video/quiz · full clickstream · storage · analytics · CSV/Excel/JSON · Git history · docs.

\+ **Extra polish**

Admin analytics scope · neo-brutalist light/dark UI · offline videos · batched beacon tracking · Moodle-exact export.

Repo includes **README**, **ARCHITECTURE.md**, this deck & a demo-video script.

## THANK YOU 🎓